Department :	Computer Science and Engineering				
Course Name & Code :	Artificial Intelligence, CSS 2203			Core ~~/ PE / OE~~	
Semester & Branch :	IV		CSE (AIML)		
Name of the faculty :	Dr Vidhya.V				
No of contact hours/week:	**L**	**T**	**P**	**C**	
	3	1	0	4	

Lesson Plan

L No	Topics	CO Addressed
0	*To address OBE philosophy towards NBA and IET accreditation, followed by a discussion on Course Plan, Course Outcomes, CO-PO Mapping, and Bloom's Taxonomy, with prerequisites for the course, if any.*	
1	What is AI? Some definitions of AI	CO1
2	Four Categories of AI and Turing test	CO1
3	History of Artificial Intelligence, The State of the Art	CO1
4	Introduction: How agents should act	CO1
5	Mapping from percept sequences to actions,	CO1
6	Agents, and Environments	CO1
7	Structure of intelligent agents: Agent programs Simple reflex agents	CO2
8	Model-based, Goal-based agents,	CO2
9	Utility-based agents and learning agents	CO2
10	Behaviour and environment in which a particular agent operates, properties of agents.	CO2
11	Problem Solving and Search Techniques: Defining the problem as a state space	CO2
12	Search State space representation, production systems	CO2
13	Problem and production system characteristics	CO2
14	Uninformed Search: Breadth First Search	CO3
15	Depth First Search	CO3
16	Depth-limited search	CO3
17	Iterative deepening DFS	CO3
18	Uniform cost search.	CO3
19	Heuristic Search Techniques: Hill Climbing and variants	CO3
20	Best First Search, A* algorithm	CO3
21	Genetic algorithm search.	CO3
22	Adversarial search: Min-Max search procedure,	CO3
23	Alpha–Beta pruning.	CO3
24	Knowledge-based agents, The Wumpus World environment	CO4
25	specifying the environment, acting, and reasoning in Wumpu's world	CO4
26	representing reasoning and logic: Logic	CO4
27	Propositional logic	CO4
28	Propositional Theorem Proving, Agents based on propositional logic	CO4
29	Representing simple facts in logic	CO4
30	Representing instances and ISA relationship	CO4
31	Compatible functions	CO4
32	Compatible predicates.	CO4
33	Ontological Engineering,knowledge representation using predicate calculus	CO5
34	Knowledge engineering process	CO5
35	Probability and Bayes' rule	CO4

36	Expert system shells, Knowledge acquisition	CO5

Faculty members teaching the course (if multiple sections exist):

Faculty	Section

References:

Textbooks	• Stuart Russell and Peter Norvig – Artificial Intelligence A Modern Approach (3e), Pearson Education, Third Edition, 2010. • Elaine Rich, Kevin Knight, Shivashankar B. Nair, Artificial Intelligence (3e), Tata McGraw Hill, 2010.
Self-Directed Learning	
Research Literature/ Case Studies	• **https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10122500** • **chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://ai.vub.ac.be/sites/default/files/sab94.pdf**
NPTEL/Coursera/any MOOC-based material	• **Artificial Intelligence (Prof.P.Dasgupta), IIT Kharagpur https://nptel.ac.in/courses/106105079** • **History of Artificial Intelligence https://www.youtube.com/watch?v=056v4OxKwlI** • **Rise of Artificial Intelligence https://www.youtube.com/watch?v=eX7iYN4wCWo** • **Knowledge Representation in AI - https://www.youtube.com/watch?v=eX7iYN4wCWo** • **Ontology Engineering - https://www.youtube.com/watch?v=FLsPcP9UPuM**